E3: 245 Processor System Design — Laboratory Exercise 3

5-Stage Pipelined 32-bit RISC-V Processor with Direct-Mapped Instruction Cache

Anirudh Chaturvedi (SR No: 25850)
DESE, Indian Institute of Science, Bangalore

November 12, 2026

1 Introduction & Design Specifications

This report documents the design, FPGA implementation, and post-route verification of a **5-stage pipelined 32-bit RISC-V (RV32I) processor** with a **direct-mapped Instruction Cache** targeting the Digilent Basys 3 FPGA board (`XC7A35T-1CPG236C`, Artix-7, speed grade `-1`).

- **Supported ISA (User Mode):** Basic RV32I arithmetic-logic instructions (R-type and I-type), `LUI`, branch instructions (`BEQ`, `BNE`, `BLT`, `BGE`, `BLTU`, `BGEU`), load/store instructions (`LB`, `LH`, `LW`, `LBU`, `LHU`, `SB`, `SH`, `SW`), and jump instructions (`JAL`, `JALR`). Atomic, `AUIPC`, and CSR instructions are omitted as specified.
- **Pipeline & Forwarding:** 5-stage pipeline (**IF, ID, EX, MEM, WB**) with full data forwarding from **EX/MEM** and **MEM/WB** outputs to the EX stage (`forwarding_unit.v`), plus **WB-to-ID** write-through bypass in the register file (`reg_file.v`).
- **Hazard Detection & Stall Unit (`hazard_unit.v`):** Detects load-use hazards (1-cycle stall of `PC` and `IF/ID`, bubble injected into `ID/EX`), resolves branches/jumps in EX (flushing `IF/ID` and `ID/EX`), and freezes all pipeline stages during Instruction Cache misses.
- **Direct-Mapped Instruction Cache (`icache.v`):** 16 cache lines × 4 words/block (64 words = 256 bytes), backed by a 512 × 32-bit synchronous **Block RAM** main instruction memory (`instr_bram.v`). On a miss, 4 words are fetched sequentially from BRAM without burst transfers.
- **Unaligned Data Memory (`data_mem.v`):** 2048-byte memory organized as 4 × 512 × 8-bit Distributed RAM banks supporting single-cycle unaligned byte, half-word, and word transfers.

2 Datapath Block Schematic

Figure 1 illustrates the 5-stage pipelined RV32I datapath, Instruction Cache hierarchy, forwarding paths, and hazard control signals.

Table 1: Direct-Mapped Instruction Cache Address Decomposition (32-bit Byte Address)

Bits [31:8] (24 bits)	**Bits [7:4] (4 bits)**	**Bits [3:2] (2 bits)**	**Bits [1:0] (2 bits)**
Tag	Cache Index (0–15)	Word Offset (0–3)	Byte Offset (`00`)

3 Controller & I-Cache FSM State Diagram

Figure 2 shows the Instruction Cache miss/fill controller FSM, and Table 2 summarizes the pipeline stall and flush truth table.

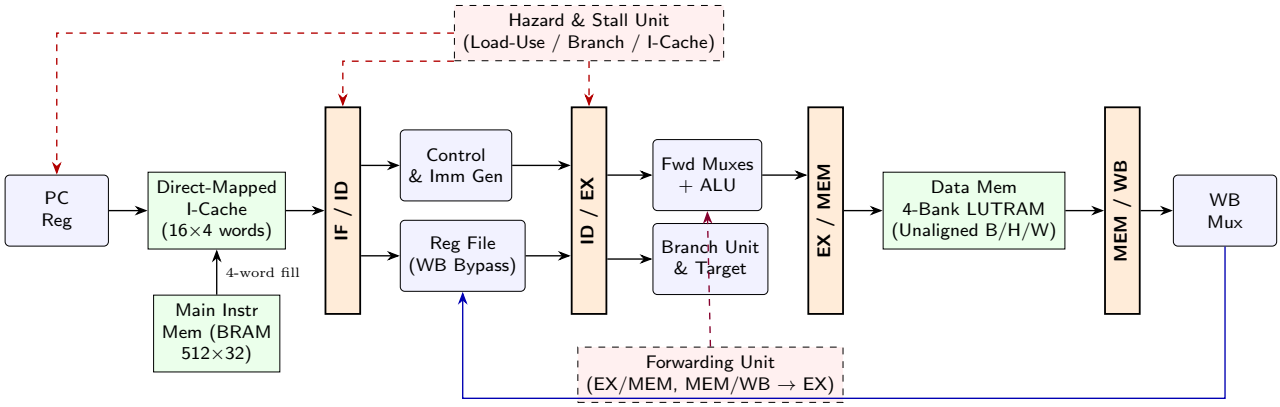


Figure 1: 5-Stage Pipelined RV32I Processor with Direct-Mapped Instruction Cache.

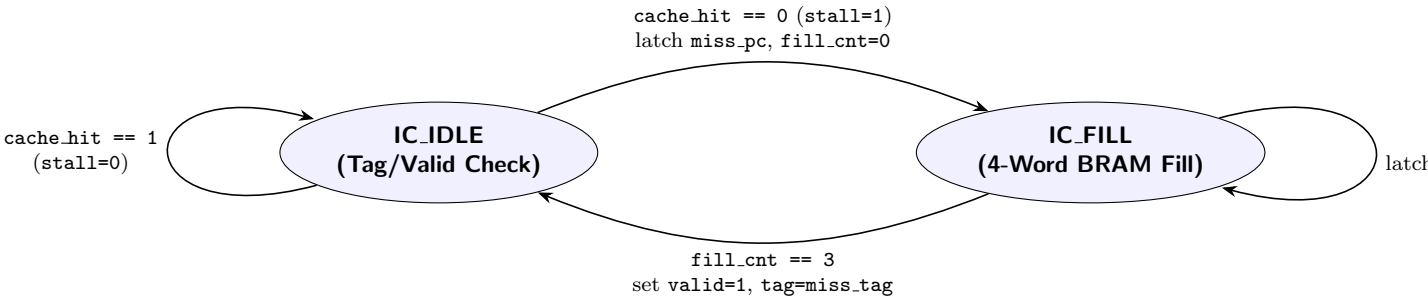


Figure 2: Instruction Cache Miss/Fill Controller FSM.

Table 2: Hazard Detection & Pipeline Stall/Flush Control Truth Table

Condition	`stall_if/id`	`stall_ex/mem/wb`	`flush_if_id`	`flush_id_ex`
`icache_stall == 1`	1	1	0	0
`ex_redirect == 1` (Branch/JAL/JALR)	0	0	1	1
`load_use_stall == 1`	1	0	0	1
Normal Execution	0	0	0	0

4 Verification: 16-bit Unsigned Restoring Division

The processor was tested with a C/RISC-V assembly program (`restoring_div.c`, `restoring_div.asm`) implementing `restoring_divide(100, 7, &rem)` as a function called via `jal ra, restoring_divide` and returning via `jalr x0, ra, 0`, followed by unaligned byte/half-word/word memory tests and a back-to-back load-use hazard.

Table 3: Behavioral (`iverilog`) and Post-Route Timing (`xsim`) Verification Results

Check / Metric	**Observed Value**	**Status**
Quotient `MEM[0x200]` (100/7)	`14` (`0x0000000E`)	PASS (Behavioral & Post-Route)
Remainder `MEM[0x204]` (100 mod 7)	`2` (`0x00000002`)	PASS (Behavioral & Post-Route)
Unaligned `MEM[0x208]` (`sb`/`sh`)	`0xC0DE5A00`	PASS
Unaligned + Load-Use `MEM[0x210]`	`0x00C0DE5B`	PASS
I-Cache Misses / Stall Cycles	10 misses / 50 stall cycles	Verified (5 cycles/miss)
Load-Use Stalls / Branch Flushes / Fwd	1 stall / 106 flushes / 73 fwd	Verified

5 Vivado 2026.1 Post-Route Timing & Resource Utilization

The complete design (`basys3_top.v`) was synthesized, placed, routed, and verified with post-implementation timing simulation in **Vivado 2026.1** for `xc7a35tcpg236-1` with a 100 MHz (`10.000 ns`) clock constraint.

Table 4: Post-Route Resource Utilization and Timing Summary (`xc7a35tcpg236-1`)

Resource / Timing Metric	**Used / Value**	**Available / Target**	**Utilization / Margin**
Slice LUTs (Total)	2,955	20,800	14.21%
LUT as Logic	2,699	20,800	12.98%
LUT as Distributed RAM (`data_mem`)	256	9,600	2.67%
Slice Registers (Flip-Flops)	4,165	41,600	10.01%
Block RAM (`RAMB18E1` for `instr_bram`)	0.5 Tile (1 RAMB18)	50 Tiles	1.00%
DSPs	0	90	0.00%
Worst Negative Slack (WNS @ 100 MHz)	−0.596 ns	10.000 ns	$F_{\max} \approx 94.37$ MHz
Worst Hold Slack (WHS)	+0.128 ns	0.000 ns	0 hold violations
Worst Pulse Width Slack (WPWS)	+3.750 ns	0.000 ns	0 pulse-width violations